

MARMARA UNIVERSITY

Faculty of Engineering — Dept. of Electrical and Electronics Engineering

Project Proposal Form

Courses: EE4084 - Kalman and Bayesian Filters**Prepared By:** Alperen Tufan Pelit — Rüzgar Batı Okay**Submission Deadline:** 22/04/2026 @ 23:55

1 Project Identity

Proposed Title: Autonomous Vehicle Tracking and State Estimation via Sensor Fusion using CARLA Simulator

2 Motivation & Problem Statement

Accurate road slope (pitch) estimation is critical for the longitudinal control of autonomous vehicles. While deep learning offers data-driven solutions, it acts as a "black box" lacking mathematical interpretability. To establish a robust, model-based framework, this project proposes fusing IMU and monocular camera data.

Since IMU data suffers from accumulating drift and cameras from visual noise, a probabilistic approach is essential. Furthermore, vehicle kinematics and camera-to-world projection geometries are highly nonlinear. Standard linear assumptions fail in such practical tracking scenarios, strictly require robust Bayesian frameworks [1]. Because basic linearization of these highly nonlinear transformations can cause severe estimation errors and filter divergence [2], employing advanced Bayesian estimation (such as EKF) is critical.

In our proposed architecture, high-frequency IMU data dynamically predicts the vehicle state, while road pitch angles, geometrically derived from camera-extracted vanishing horizon points, provide iterative measurement updates to correct the drift. The CARLA simulator will supply both realistic noisy sensor measurements and absolute ground-truth data via GPS sensor for precise quantitative validation.

3 Mathematical Framework

3.1 Process and Measurement Models

The state vector x_k describes the vehicle's road pitch angle θ and pitch rate $\dot{\theta}$ at time step k :

$$x_k = [\theta \quad \dot{\theta}]^T \quad (1)$$

- **Evolution:** The state evolves according to a kinematic model where high-frequency IMU data acts as the control input u_{k-1} for the prediction step:

$$x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}, \quad w_k \sim \mathcal{N}(0, Q) \quad (2)$$

- **Observation:** The measurement z_k is the pitch angle extracted geometrically from the vanishing point of lane lines in the camera frame:

$$z_k = h(x_k) + v_k, \quad v_k \sim \mathcal{N}(0, R) \quad (3)$$

3.2 Extended Kalman Filter Recursion

Since $f(\cdot)$ and $h(\cdot)$ are nonlinear, the EKF linearizes them using the Jacobian $F_k = \frac{\partial f}{\partial x} \big|_{x=\hat{x}_{k-1|k-1}}$ at each step. The filter runs in two steps:

Predict:

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_{k-1}) \quad (4)$$

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^\top + Q \quad (5)$$

Update:

$$K_k = P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + R)^{-1} \quad (6)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (z_k - h(\hat{x}_{k|k-1})) \quad (7)$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} \quad (8)$$

where $H_k = \frac{\partial h}{\partial x} \big|_{x=\hat{x}_{k|k-1}}$ is the observation Jacobian and K_k is the Kalman Gain.

4 Data Acquisition and Generation Strategy

- **Ground Truth Generation:** True vehicle pitch states $x_{0:K}$ are taken directly from CARLA. The simulator outputs precise rotation data at each time step through its Python API to be used as the reference.
- **Measurement Simulation:** Camera measurements $z_{1:K}$ (pitch angle) and IMU control inputs $u_{1:K}$ (pitch rate) are created by adding Gaussian noise to the true CARLA sensor outputs. The camera sampling frequency is $f_s = 20$ Hz with a target SNR of 20 dB. Visual outliers and random signal dropouts are added to the camera data to simulate real-world conditions such as temporary lane occlusions or lighting glare.
- **External Datasets:** No external dataset is needed. CARLA provides all required high-fidelity data to simulate both IMU and monocular camera streams for road pitch estimation.

5 Algorithmic Approach & Software Strategy

Filter Choice & Explanation:

We employ the Extended Kalman Filter (EKF). The system's IMU-driven kinematics and camera-based vanishing point geometry are inherently nonlinear. While UKF or Particle Filters handle non-linearities well, they add unnecessary computational overhead for our focused, low-dimensional state vector ($x_k = [\theta, \dot{\theta}]^\top$). The EKF provides the optimal balance, ensuring real-time efficiency for sensor fusion via Jacobian linearization without sacrificing mathematical rigor.

Software Strategy & Libraries:

The project will be developed in Python to integrate with the simulation environment. Primary tools are as follows:

- **CARLA Python API:** For environment setup and acquiring asynchronous IMU, camera, and absolute ground-truth data.
- **NumPy:** To execute fast matrix operations, compute Jacobians, and manage EKF state updates.
- **OpenCV:** To perform traditional computer vision tasks (e.g., edge detection, Hough transforms) to extract the vanishing point geometrically, strictly avoiding deep learning models.
- **Matplotlib:** For visualizing filter convergence, IMU drift correction, and computing the RMSE against the ground truth.

6 Milestones

Week	Primary Objective
W1-2	Mathematical derivation of the IMU-kinematic and Camera-geometric models, and formulation of EKF Jacobians.
W3-5	CARLA environment setup, OpenCV vanishing point algorithm development, and core EKF Python implementation.
W6-7	Monte Carlo analysis (RMSE/NEES), sensor dropout simulations, and final reporting.

Instructor Approval

Prof. Dr. Engin Masazade

References

- [1] M. Arulampalam, S. Maskell, N. Gordon, and T. Clapp, “A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking,” *IEEE Transactions on Signal Processing*, vol. 50, no. 2, pp. 174–188, 2002.
- [2] S. Julier and J. Uhlmann, “Unscented filtering and nonlinear estimation,” *Proceedings of the IEEE*, vol. 92, no. 3, pp. 401–422, 2004.